

The background of the slide is a dark blue gradient with a complex network of thin, light blue lines connecting various points. These points are represented by small circles in shades of blue and white, creating a web-like or molecular structure that spans the entire frame.

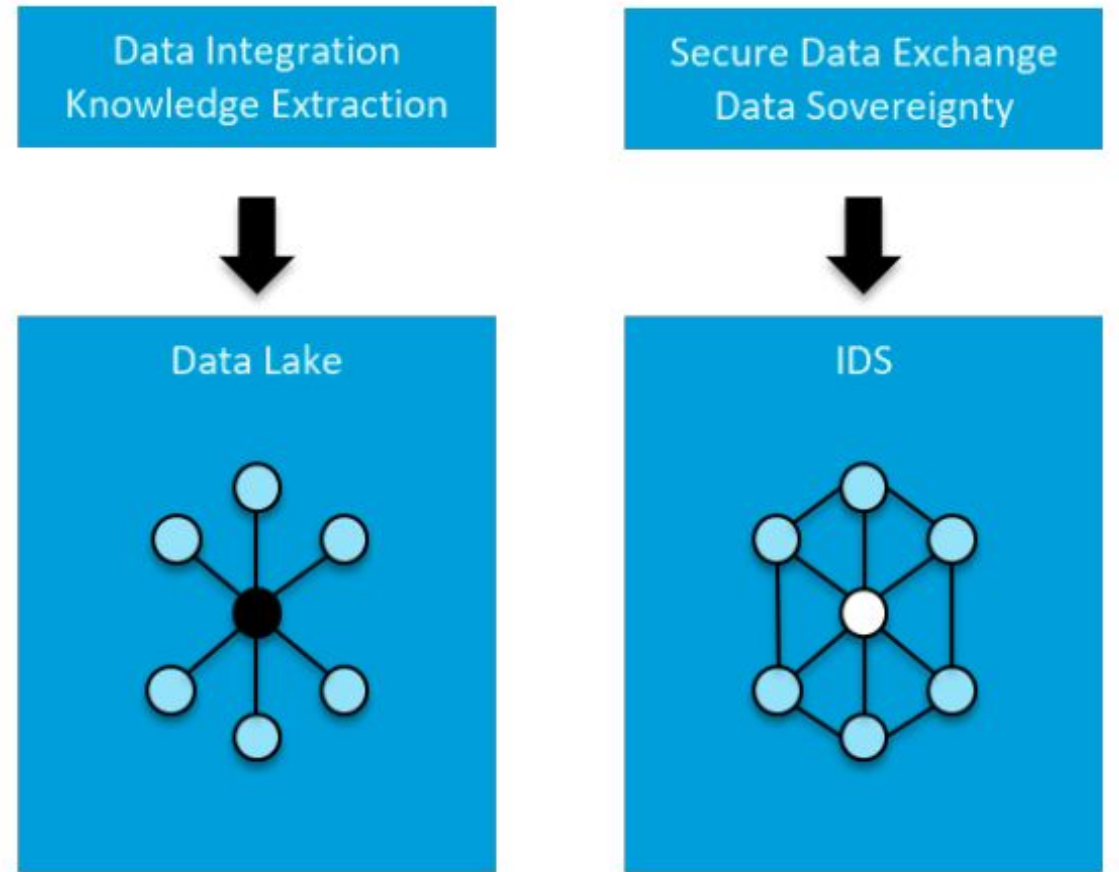
OneNet Data Space

LFE TAC Meeting – Sept 8th

Data Space and OneNet Concepts

Data Space

- Data Spaces offer to data driven Ecosystems a **secure, decentralized data sharing by design**. A dataspace allows the participants within the federated network to **maintain control over their data, boosting their sovereignty**.
- Unlike data integration over traditional centralized Data systems, a dataspace system does not have full control on its data and gradually integrates data as necessary.
- The actual goal of a data space is to provide base functionality over all data sources, regardless of how integrated they are, enhancing the functional and semantic interoperability across different platforms and industries, offering commercial and functional advantages among the participants.



Building Blocks

Data Privacy
and Security

Trust and security among Data Space participants

Data Storage
and Access
Control

Decentralized Storage and Secure Access/Usage Control

Interoperability
and Standards

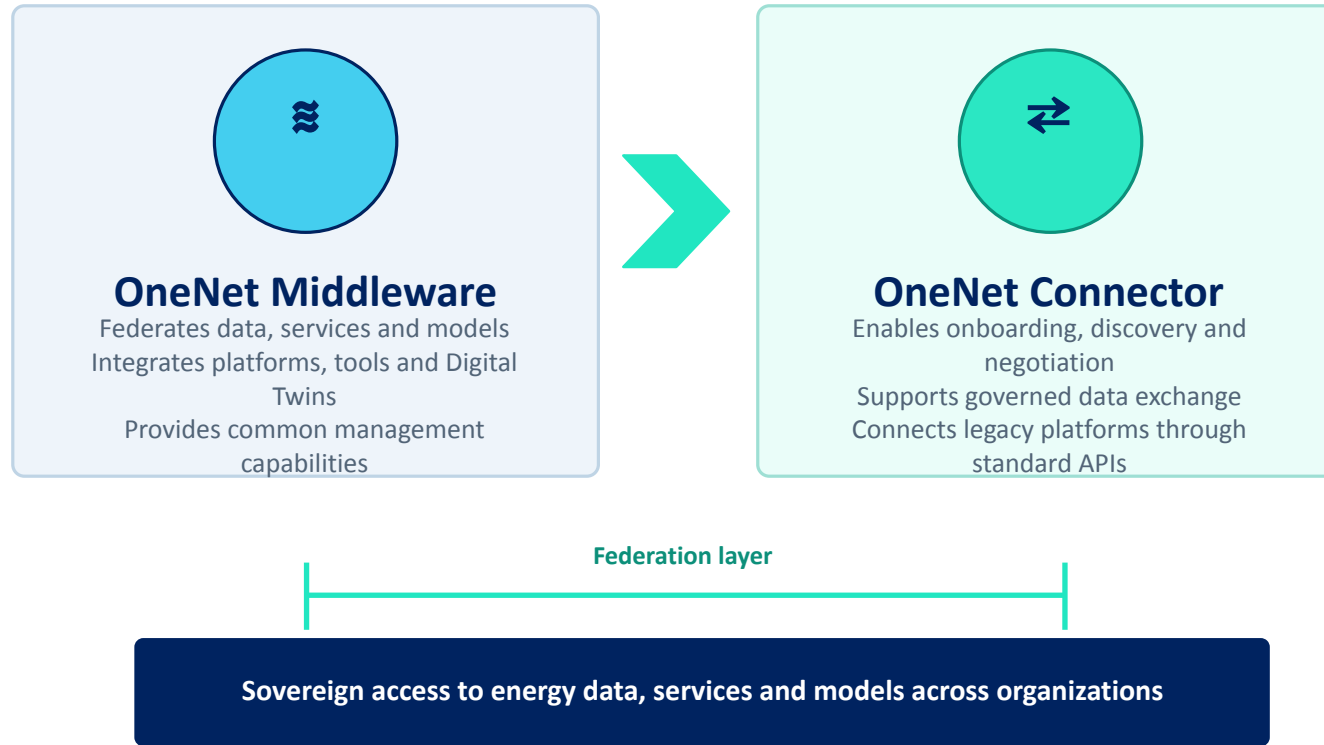
Technical and Semantical interoperability for seamless data exchange

Integration and
Data Exchange

Federated Data Catalogue and end-to-end Data Exchange

OneNet Data Space: from technology to a sustainable ecosystem

ONE DATA SPACE, TWO COMPLEMENTARY BUILDING BLOCKS



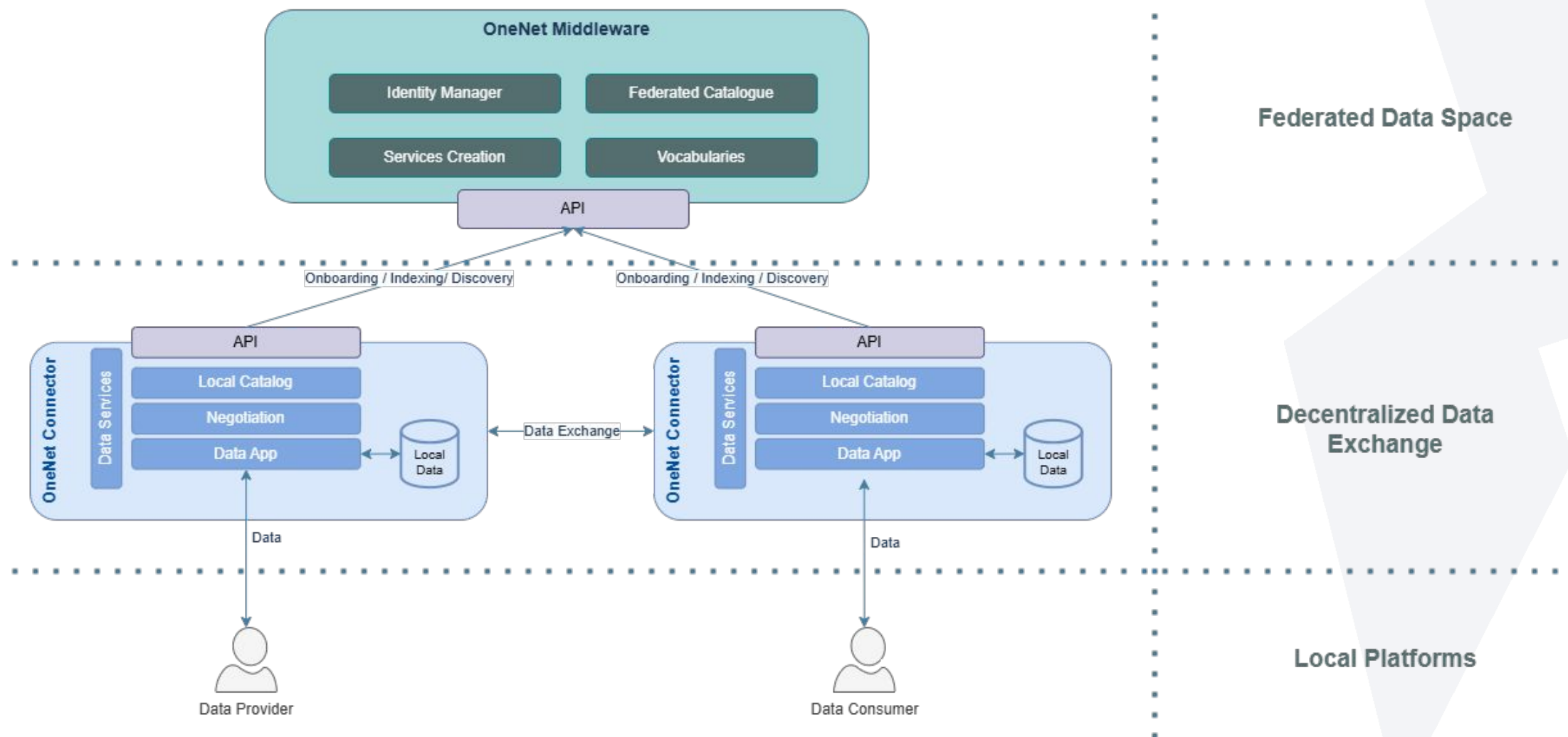
WHY LINUX FOUNDATION?

Turn a project asset into a trusted, industry-wide open ecosystem

- 1 Neutral governance**
A trusted home where no single organization controls the project
- 2 IP clarity & trust**
Clear contribution and asset-management frameworks reduce uncertainty
- 3 Community & adoption**
Broader participation accelerates contributions, use cases and market uptake
- 4 Long-term sustainability**
Shared stewardship supports maintenance, evolution and commercial engagement

Strategic outcome: establish OneNet as a reusable open-source reference implementation for interoperable and sovereign energy data sharing.

Decentralized Data Space



The background features a complex network diagram with numerous nodes and connecting lines, rendered in a light blue color against a dark blue gradient background. The nodes are represented by small circles, and the lines are thin, creating a web-like structure that spans the entire frame.

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OneNet Middleware

OneNet Middleware in the Data Space ecosystem

The OneNet middleware: A scalable, unified platform for federated data, service and model sharing

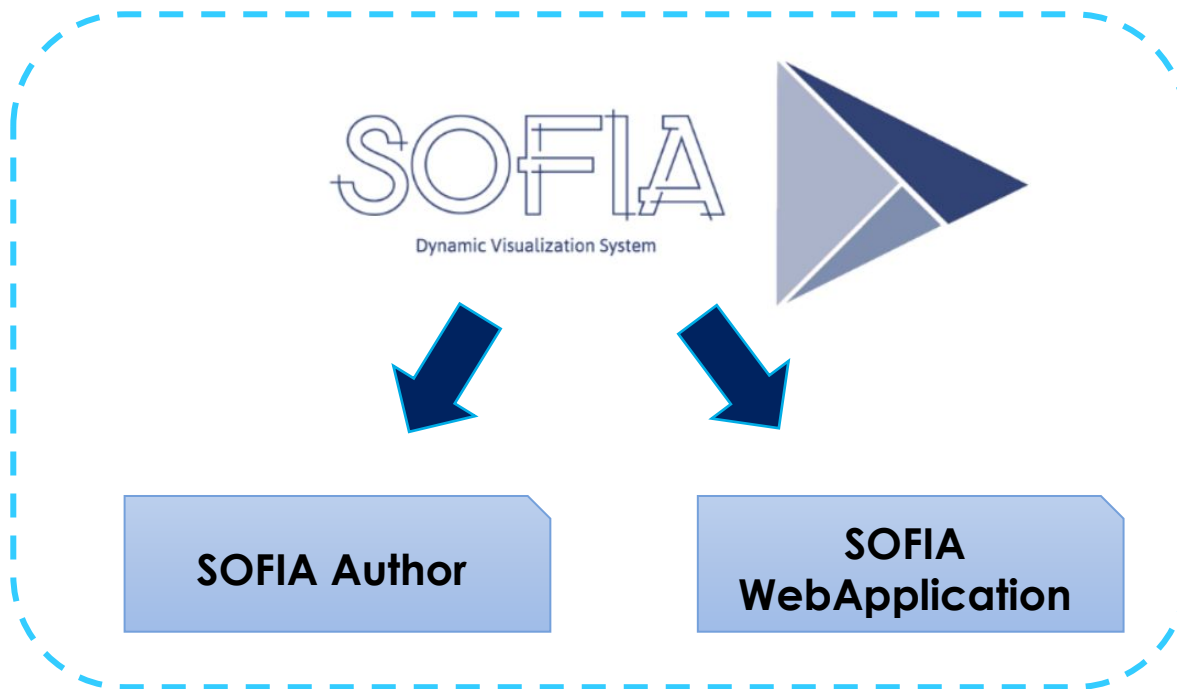
Participants Onboarding
It allow to create a Data Space ecosystem with standardized identification and authorization processes

Data Management
The middleware supports managing data, models, and resources in a federated way, ensuring alignment with Data space reference architecture models

Service integration
Enables access to external data and services for energy stakeholders in a secure and trusted way

OneNet Middleware development

OneNet Middleware development is based on European Dynamic's Web Application Generator platform **SOFIA**



SOFIA comprises of two sub-modules: The **Author Application** which is used to dynamically configure the presentation of the data, the menus, graphs, and user access and the **Web Application** which visualizes the data in various configurations and graphics. It is built on **JAVA Spring boot, Angular, Redis, MySQL** Databases and **Docker** technologies.

➔ **OneNet Connector API and Graphical User Interface**

The background of the slide is a dark blue gradient with a complex, abstract network of thin, light blue lines connecting numerous small, semi-transparent blue dots. These dots and lines are scattered across the entire frame, creating a sense of a global or interconnected digital space. The lines vary in opacity and thickness, and the dots are also semi-transparent, allowing the background to show through them.

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OneNet Connector

The Data Space enabler

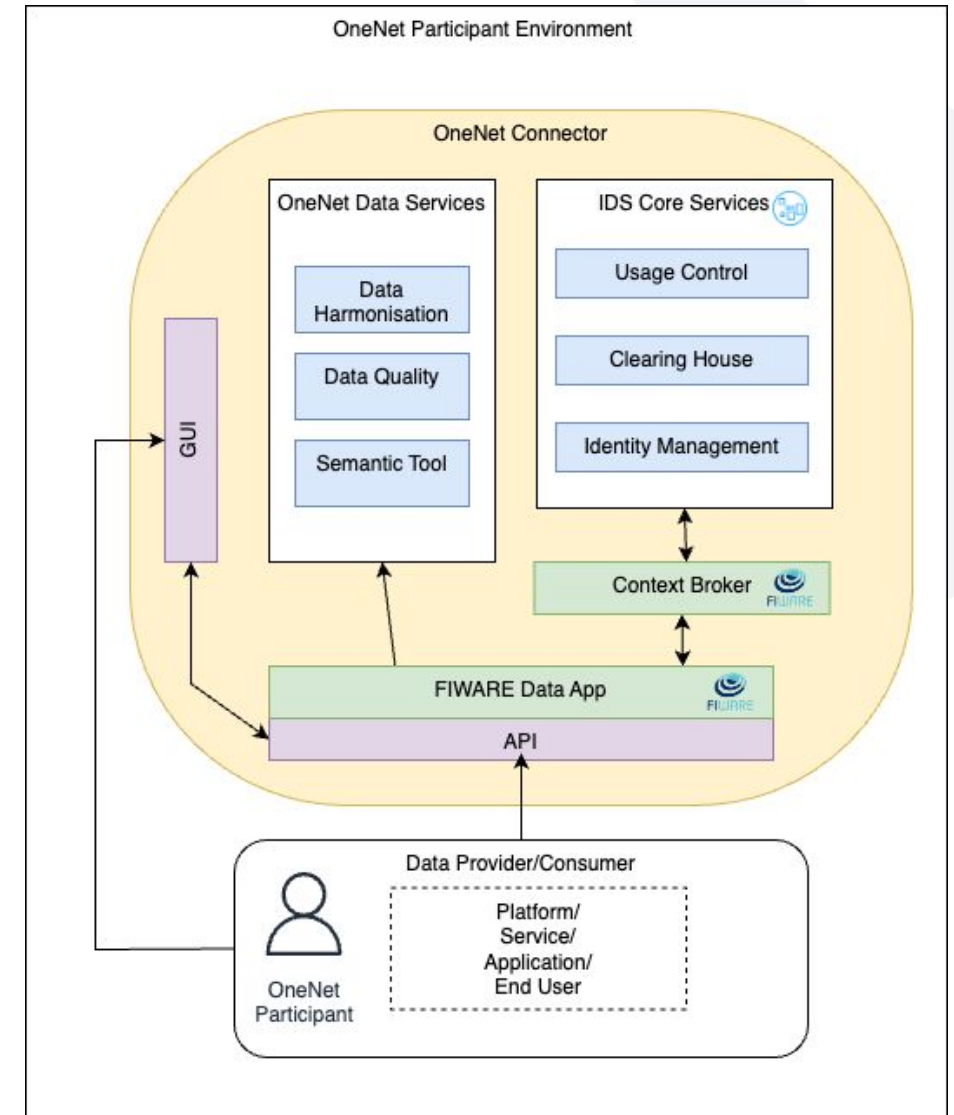
Starting Point: OneNet

Main Characteristics

- + FIWARE Context Broker fully integrated in the Connector Architecture
- + Additional Data Services (e.g., CIM Data Harmonization Tool)
- + Standardized Interfaces
- + Open GUI for facilitating the Connector usage

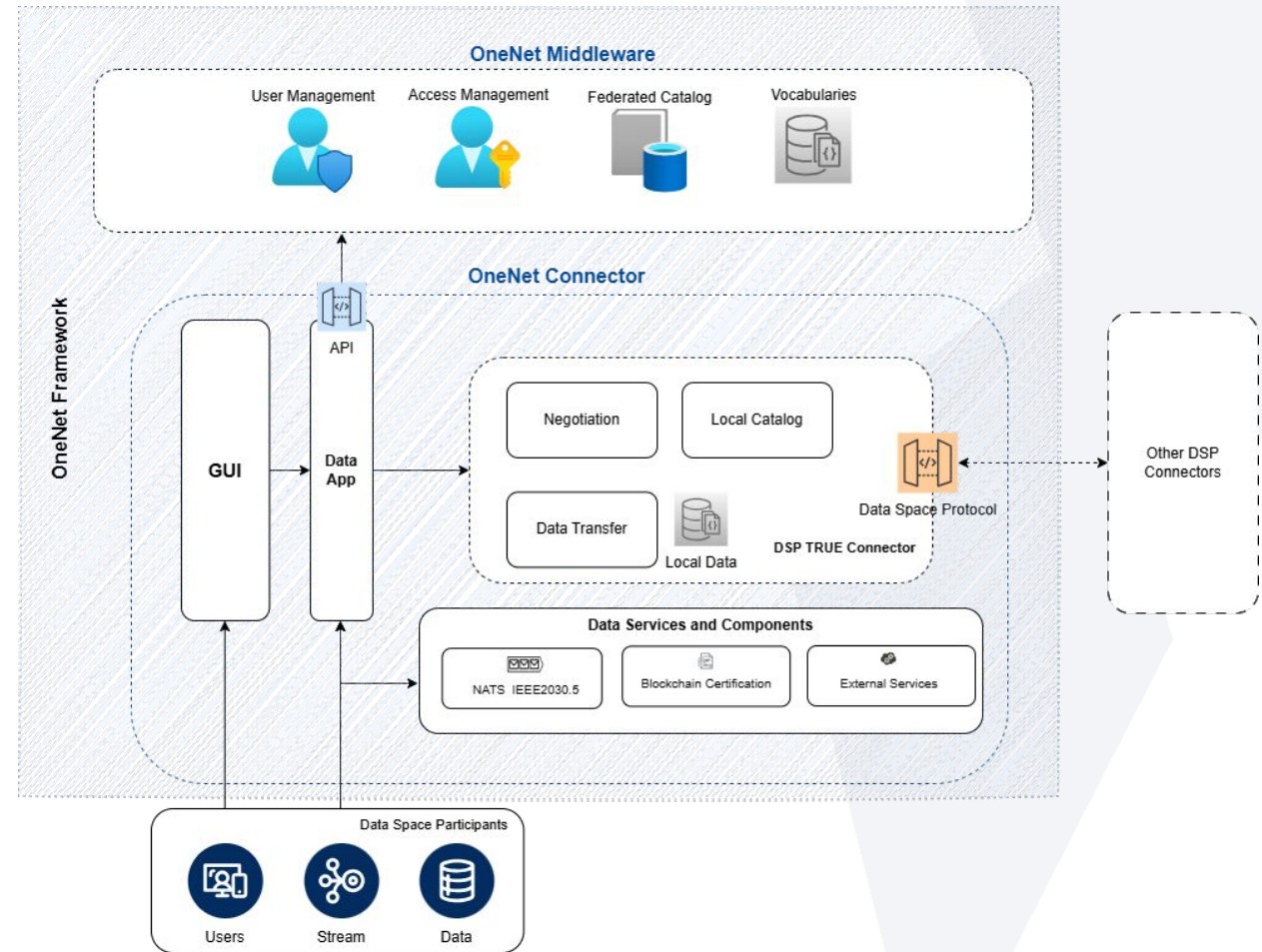
The energy context... Tailoring for Energy Domain

- + 10 services categories
- + 64 energy services
- + Vocabularies adoption



Current Version

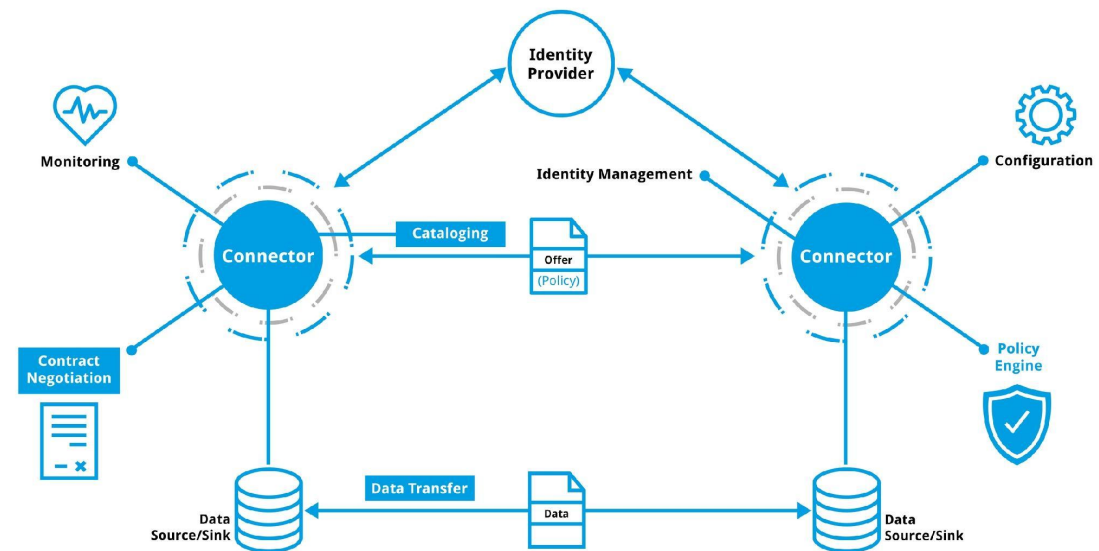
- + Fully compliant with Data Space Protocol 2024-1
- + Open API and Data App (standardized interfaces for integrating legacy platforms and Middleware)
- + Graphical User Interface (GUI) for configuration and main functionalities



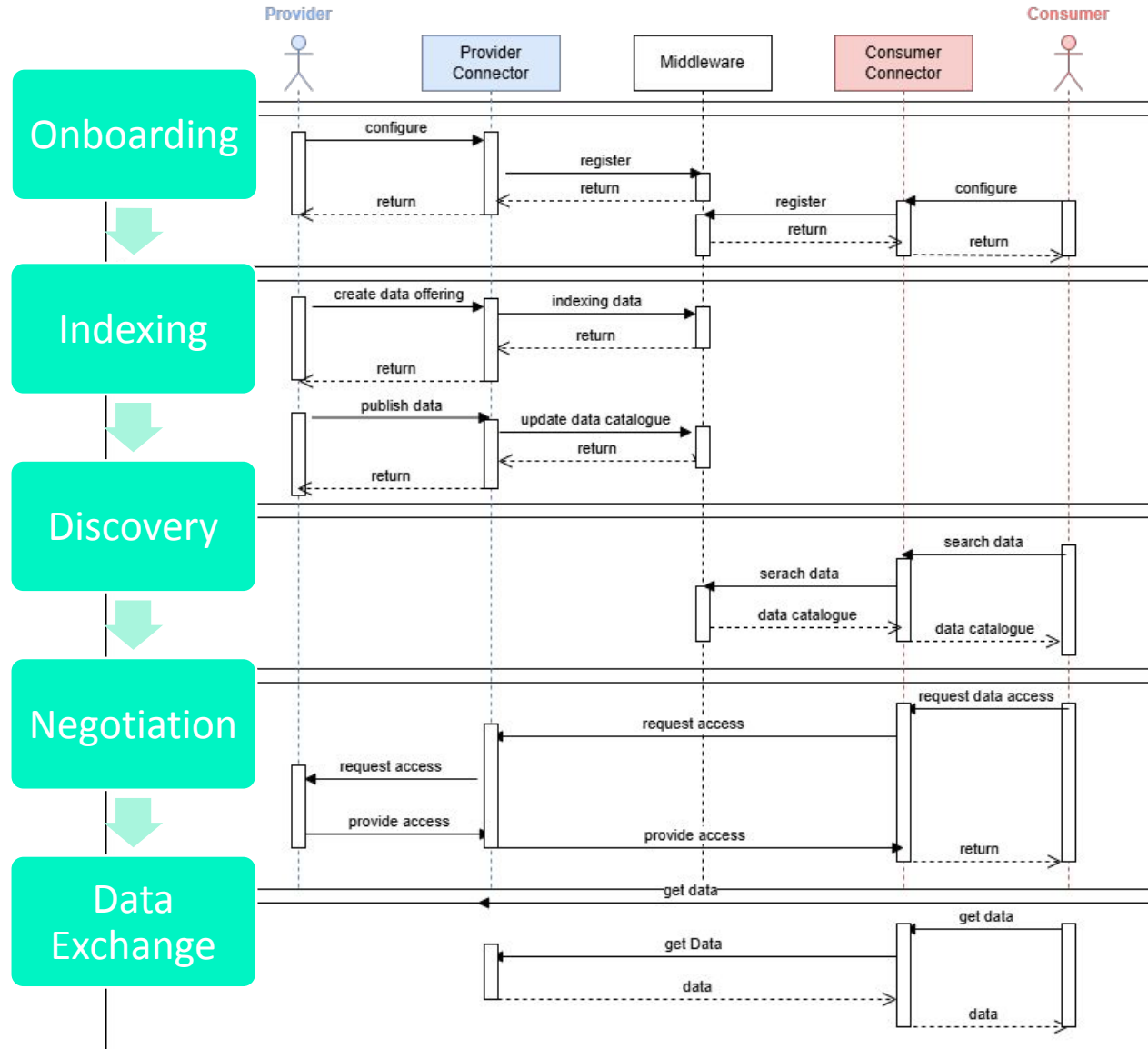
Data Space Protocol

The **Dataspace Protocol** is a set of specifications designed to facilitate interoperable data sharing between entities governed by usage control and based on Web technologies. These specifications define the schemas and protocols required for entities to publish data, negotiate Agreements, and access data as part of a federation of technical systems termed a Dataspace.

The Dataspace Protocol is used in the context of Dataspaces with the purpose to support **technical interoperability** across **Data Spaces**



Main processes



Advanced GUI

The image displays two screenshots of the OneNet DSP Connector user interface, showing an advanced GUI for managing data services and subscriptions.

Top Screenshot: My Offered Services

The top screenshot shows the "My Offered Services" page. The left sidebar contains navigation options: Dashboard, Catalog, Services (selected), Service Categories, My offered services, and Requests. The main content area displays a table of offered services with columns: #, Category, Title, Created On, Profile Format, Profile Description, Status, and Subscriptions. A "New Data Service +" button is visible in the top right corner.

#	Category	Title	Created On	Profile Format	Profile Description	Status	Subscriptions
1	Forecasts	Forecasts Dynamic line rating	Test Data Offering 2025-07-11T15:27:25.941Z	11/07/2025 15:27		active	Engineering Ingegneria Informatica userTest2

Bottom Screenshot: Dashboard

The bottom screenshot shows the "Dashboard" page. The left sidebar contains navigation options: Dashboard (selected), Catalog, Services, Data exchange, and Connector settings. The main content area displays several status cards:

- Your Data Entities Consumed:** 3 Total. Go to Data Consumed List.
- Your Data Entities Provided:** 123 Total. Go to Data Provided List.
- Your Offered Services:** 69 Total. Go To Offered Services List.
- Your Subscriptions to Services:** 1/1 Accepted/Total. Go To Subscriptions List.
- Pending Subscription Requests to Your Offers:** 5 Pending. Pending Subscription Requests to Your Offers.
- Connector Check:** Checking connections please wait... Go To Your Connector Settings.

Additional Features

- + **Integration of external services (Push Mechanism)**

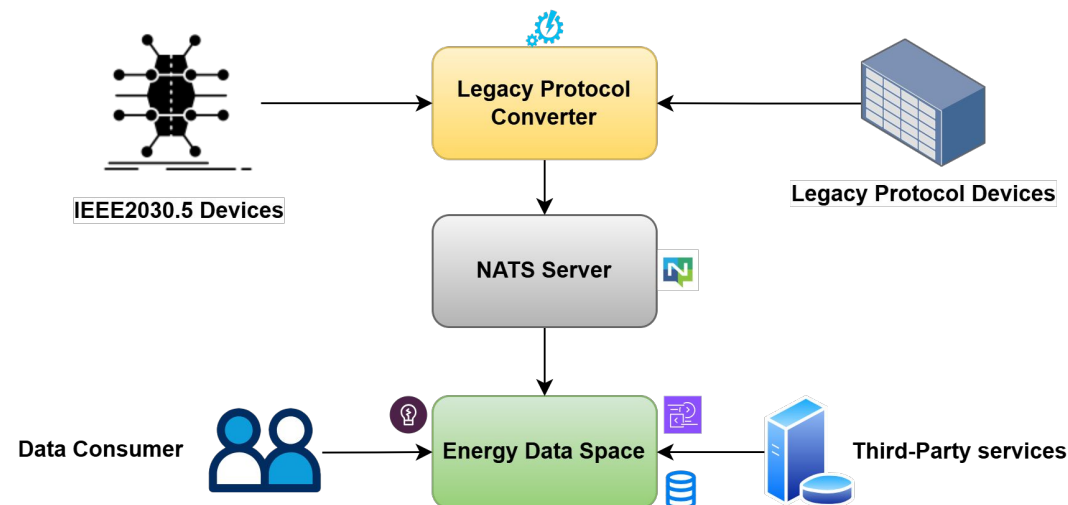
- + Provide data to external service in a (near) real-time way, maintaining the features related to the Data Space

- + **Blockchain Transaction Notarization**

- + Logging and Certification of data transaction in an immutable and secure way

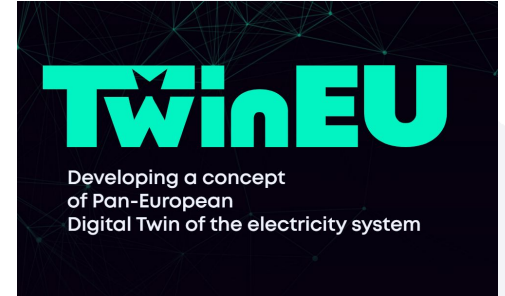
- + **New Standards (IEEE2030.5) and real-time communication protocols (NATS)**

- + Integration of new standards and protocols for a wider support



OneNet Ecosystem

- The project **TwinEU** is coming soon to an end but the work will continue in other EU projects
- An important asset is the development of the **INSIEME** project that is moving the Common European Energy Data Space **CEEDS** from the research environment to the industrial application
- The EU Commission is also driving a **standardization Task Force** for data protocols and exchanges where the OneNet connector is supposed to play a key role
- **ENG** is the **Sector Orchestrator for Energy Domain in Data Spaces Support Centre 2 (DSSC2 2026-2029)**.
 - This initiative focusses on **concreteness** and **maturity for market sustainability** on respect of DSSC that has been focused on the creation of technological blueprints, interoperability standards and governance guidelines
 - It is planned to have **OneNet Maturity Assessment** and participate to the second round of **European DataSpace Awards**
- These development should bring in the close future **more participant to the Community**



Thank you!



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